GEO Meta-Analysis Toolkit

Autonomous Multi-Tool Agent for NCBI GEO Exploration, Metadata Aggregation, & AI Narrative Landscape Synthesis

⚡ 3-Phase Pipeline | 🧠 Local Ollama medgemma:4b | 🔬 NCBI Entrez E-Utils | 🧬 FAISS Vector Index

THE CHALLENGE

Biomedical Discovery Bottlenecks

Why traditional manual search and generic linear scripts fall short in high-throughput genomics.

📦 Data Scale & Complexity

- NCBI GEO holds >160,000 datasets across thousands of organisms.
- Complex search syntax (Entrez Boolean & Field tags).
- Non-standardized sample descriptions and metadata schemas.

⏳ Manual Synthesis Time

- Extracting organism distributions and timelines takes hours.
- Manual cross-referencing of PubMed literature.
- Prone to human error when handling hundreds of GSE records.

🤖 LLM & Hardware Specs

- Pure agentic ReAct loops stall on LLM reasoning timeouts.
- No Rate limits and API cost overruns.
- Laptops with 8 GB GPU VRAM.

THE SOLUTION

Unified AI-Powered Meta-Analysis

Bridging deterministic high-speed computation with autonomous agent reasoning.

< 2s

INSTANT AGGREGATION

Phase 1 direct parsing provides real-time chart rendering before LLM starts.

100%

EXECUTION RESILIENCE

3-phase execution ensures no search query ever fails.

3.5 GB

VRAM EFFICIENCY

Tailored for local GPU acceleration using bio LLM medgemma:4b.

10,000

MAX SCALABILITY

Scalable data fetching engine tuned with NCBI API key rate-limiting.

SYSTEM ARCHITECTURE

Unified Smart Hybrid Architecture

Modular design decoupling user interaction, agent reasoning, API tools, and semantic RAG vector search.

Streamlit Dark UI Dashboard

Summary Metrics • Visual Charts • Datasets Table • Reasoning Trace

Unified Smart Hybrid Engine

Phase 1: Direct GEO E-Utils → Phase 2: LangChain ReAct PubMed → Phase 3: Auto Fallback

Local GPU LLM

Ollama (medgemma:4b)
Strict Local GPU/CPU Execution

NCBI E-Utilities

GEO Datasets Search

PubMed Abstract Retrieval

FAISS Vector Store

Cell Ontology (CL, UBERON, EFO)

Two-Pass LLM Normalizer

EXECUTION STRATEGY

3-Phase Workflow

Eliminating LLM single-point failures through layered execution fallback.

Phase 1: Direct Engine

⚡ Execution Time: ~2 seconds

• Queries NCBI GEO directly via E-utilities API.

• Parses XML metadata instantly.

• Extracts organism counts, timelines, sample stats, and title terms.

Phase 2: ReAct AI Synthesis

🧠 Execution Time: ~10-15 seconds

• Spawns specialized LangChain ReAct Agent.

• Autonomously queries PubMed for literature context.

• Synthesizes qualitative research landscape summary.

Phase 3: Automatic Fallback

🛡️ 100% Uptime Shield

• Catches LLM timeouts, parsing errors, or network issues.

• Seamlessly auto-degrades to linear summarizer.

• Guarantees zero empty outputs on any search query.

SEMANTIC INTELLIGENCE

Ontology Disambiguation & RAG

Standardizing fragmented biomedical jargon using FAISS vector search & ontology mappings.

📚 Supported Biomedical Ontologies

- Cell Ontology (CL): Standardized cell types (e.g., CD8+ T cell, Podocyte, Microglia).
- UBERON: Anatomical structures and tissue locations across species.
- Experimental Factor Ontology (EFO): Experimental assay types, variables, and platform technology.

💡 Two-Pass Cost-Gated Disambiguation

1. Fast Vector Search: Cosine similarity lookup in pre-computed FAISS index.

2. Cost-Gated LLM Normalizer: Triggers ONLY when similarity is below 0.70 threshold.

- Resolves ambiguous jargon (e.g., 'MACS' cell separation method vs. 'macs' slang for macrophages).
- Zero extra LLM round-trips for clean terms.

INTERACTIVE ANALYTICS

Multi-Tab Meta-Analysis Dashboard

Live aggregated analytics across NCBI GEO dataset search results.

📝 Summary Tab

- Key summary metrics (total datasets, total sample volume, top organism).
- Flowing LLM narrative overview of the research landscape.

📈 Visual Charts Tab

- Organism distribution donut chart.
- Yearly publication timeline & top title keyword term frequency.

📋 Datasets Table

- Full sortable results matrix with GSE ID, Title, Organism, Sample count, and Submission date.

🔧 Raw Data & Trace

- Raw JSON response inspection from NCBI E-Utilities API calls.
- Live reasoning trace panel showing agent scratchpad & tool outputs.

HARDWARE & PERFORMANCE

Local GPU & Hardware Optimization

High-efficiency local LLM inference configured for modern AI PCs.

💻 Workstation Specs

- GPU: Minimum 8 GB VRAM
- RAM: 8 - 32 GB
- OS: Ubuntu 24.04 LTS (CUDA 12)

🧠 Quantized LLM Profiles

- medgemma:4b (~3.5 GB VRAM)
  Default bio-focused model
- gemma2:9b (~5.5 GB VRAM)
  High-reasoning upgrade
- llama3.1:8b (~4.9 GB VRAM)
  General purpose alternative

⚡ Batching & Limits

- Embedding batch size: 512 (prevents CUDA OOM).
- NCBI E-utils API: 10 req/s (with free API key).
- Max dataset limit: Scalable up to 10,000 results per query.

DEVELOPER GUIDE

Quick Start & Setup Workflow

Simple 5-step terminal deployment for local development and research.

Terminal Deployment Commands

```
# 1. One-time setup script (CUDA, Python venv, dependencies)
chmod +x scripts/setup_ubuntu.sh
./scripts/setup_ubuntu.sh

# 2. Configure environment (.env)
cp .env.example .env # Set NCBI_EMAIL='your_email@example.com'

# 3. Pull local GPU LLM model
ollama pull medgemma:4b

# 4. Build semantic ontology vector index
python scripts/build_index.py

# 5. Launch the interactive dashboard
streamlit run app.py
```

ROADMAP & CONCLUSION

Future Vision & Summary

Empowering bioinformaticians with autonomous research intelligence.

🚀 Key Achievements

✓ 100% resilient 3-phase hybrid execution flow.

✓ Sub-second statistics aggregation for up to 10,000 datasets.

✓ Offline local GPU AI inference with medgemma:4b.

✓ FAISS RAG semantic term normalization.

🔮 Future Development Roadmap

→ Spatial Transcriptomics & Single-Cell h5ad file parsing.

→ Multi-Agent peer review consensus reports.

→ Automated differential gene expression cross-comparison.

→ PDF / LaTeX automated report generation.

Thank You! — Questions & Discussion